

KARUKKUVEL RAJ D

AVIONICS SYSTEM ENGINEER



+91 9790899854



Chennai, India



d.karukkuvelraj@gmail.com



PROFESSIONAL SUMMARY

Avionics engineer with experience in aircraft electrical systems, UAV flight control systems, and systems engineering. Skilled in requirements engineering (HLR/LLR), aircraft power architecture, power budgeting, interface definition, and system integration. Experienced in electrical load analysis, verification & validation (V&V), and MBSE methodologies using Capella/Arcadia. Familiar with flight-critical systems, electrical distribution concepts, and cross-functional engineering collaboration.

EXPERIENCE

2025 - Present

Vijigi Intellect Pvt Ltd

Avionics System Engineer

- Captured HLR/LLR for landing gear systems ensuring requirement traceability and system consistency
- Contributed to aircraft power architecture definition and electrical system integration activities
- Performed power budgeting and electrical load analysis across aircraft subsystems
- Authored engine interface architecture and supported interface management activities
- Collaborated with cross-functional teams on system architecture and integration tasks

2024 - 2025

Killis Bird LLP

Flight Controls Engineer

- Designed UAV flight controller hardware and integrated PX4 autopilot software for VTOL, fixed-wing, and multirotor UAV platforms
- Performed requirements engineering, interface definition, and system integration activities for UAV systems
- Modeled system architectures using Capella/Arcadia MBSE methodologies and supported verification & validation activities

EDUCATION

2022 - 2024

Anna University

Master of Engineering (Avionics)

CGPA:8.49

2018 - 2022

Apollo Engineering
College

Bachelor of Engineering (Aeronautical)

CGPA:8.5

SKILLS

- C++, Python, Ada
- System Architecture
- Requirements Engineering
- Interface Definition
- Model Based System Engineering
- Verification & Validation

AEROSPACE STANDARDS

- ARP4754B
- ARINC429
- AFDX/ARINC664
- DO178C

ACHIEVEMENTS

- Cleared INCOSE SEP knowledge Exam
- Qualified GATE Aerospace 2021 with AIR 665

CERTIFICATIONS

- Requirements Writing by University of South Wales
- Development of Realtime Embedded Systems by eit Digital